

Bulk nano-heterointerface secures molecular contacts in perovskite solar cells

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The development of molecule-based selective contacts has boosted the power conversion efficiencies of inverted perovskite solar cells. However, these molecular films, often assembled as monolayer or multiple layers on the substrate, are prone to molecular desorption and structural deformation, limiting the long-term stability of devices. This instability, in essence, originates from the weak contacting structure between the transparent conductive oxide and molecular layer, with a limited interface offering insufficient adhering forces to immobilize the molecules. A general architectural strategy that circumvents this fundamental limitation without compromising electronic functionality is highly demanded, but remains underexplored. We now report a universal architecture of a bulk nano-heterointerface that reconstructed the molecule-based selective layer. The substantially increased chemical interface and strengthened binding force between the molecules and rationally designed nanoscale scaffolds greatly improved the device operational stability, achieving high efficiency. The strategy proved versatile, successfully applied to various molecular systems to enhance device performances, and remained effective in upscaled devices produced via scalable blade coating.

Molecule-based selective contacts have recently emerged as a crucial component in making perovskite solar cells with high power conversion efficiencies (PCEs)^{1–6}. These molecules, featuring an oxyacid group tethered by a conjugation unit, can self-assemble into monolayer or multiple layers on the surface of a transparent metal oxide substrate such as indium tin oxide (ITO)^{7–11}. The as-formed molecular thin film, residing between the substrate and the perovskite layer, exhibits exceptional properties such as minimal parasitic absorption, favourable interfacial band alignment, high charge selectivity and efficient transporting capability^{6,12–16}. Therefore, this architectural design of charge-selective contacts has been recognized as one of the most successful approaches to achieve highly efficient inverted perovskite solar cells.

However, during the prolonged operation of perovskite solar cells, some molecules that adhere to the surface of the substrate through weak physical absorption are prone to desorb, especially under external stresses such as light, heat and mechanical loading^{2,17–22}. The desorption and the subsequent uncontrolled migration of molecules can perturb the interfacial structure and impair charge carrier dynamics, particularly when the molecular film is considered to have a merely monolayer thickness. When molecules are considered to assemble into multiple layers, those in the upper layers can hardly bind with the substrate, relying solely on weak intermolecular interactions such as van der Waals interaction for stacking^{6,23–26}. This configuration further undermines the stability of the layer, as the molecules are even more

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prone to displacement due to the lack of chemical interface to fasten and secure them. This ultimately results in structural deformation and even collapse of the intended molecular stacking architecture, thereby deteriorating the PCEs of devices¹⁹. These instability issues of molecule-based selective contacts can be inherently traced back to the fragile contact structure between the transparent conductive oxide and the molecular layers, which provides limited contact area and insufficient anchoring capability to robustly accommodate and immobilize all the molecules. To further boost the longevity of perovskite solar cells, a more robust and universally applicable architectural design that moves beyond the current framework without impairing the electronic properties is urgently needed, yet this remains largely underexplored.

Here we report a universal architectural design of bulk nano-heterointerface for molecule-based selective contacts, which reconstructed the assembly structures of molecules. This design served as a more stable and universally applicable alternative to the conventional contact mode for a molecular hole-selective layer. Instead of onto the ITO surface, the molecules were strongly assembled onto the surface of spherical silicon oxide (SiO_x) nanoparticles (NPs), which delivered significantly increased interfacial contact area, to form a robust bulk heterostructure when processed into bulk films. Owing to the dramatically increased surface-to-volume ratio in nanostructures and the strong interfacial binding, the molecules could be securely fastened, leading to greatly improved structural stability without compromising the charge-transporting behaviours. The perovskite solar cells utilizing this bulk nano-heterointerface showed a high PCE of up to 26.5% with considerably improved stability. The target device retained over 98% of its initial PCE after operating in maximum power point tracking (MPPT) at 60 °C for ~2,500 h, and retained over 99% of its initial PCE after operating in the open-circuit condition at 65 °C for ~2,000 h under continuous one-sun illumination. This nanostructured interface particularly improved the device performance in the context of scalable fabrication. The small-area target cell fabricated through the blade-coating method exhibited a high PCE of up to 25.1%. When scaled up to a blade-coated solar module with an aperture area of 32.5 cm², a substantial improvement in aperture PCE from 18.1% to 23.0% was achieved.

Constructing bulk nano-heterointerface

We incorporated the custom-synthesized (2-(pyren-1-yl)ethyl)phosphonic acid (Py3; Supplementary Fig. 1 shows the molecular structure and synthesis routes), into the dispersion of SiO_x NPs to initiate the molecular assembly. The Py3 solution was thoroughly stirred before mixing with SiO_x NPs. The mixture was then stirred for 30 min to ensure the complete assembly of molecules onto the NP surface (Supplementary Fig. 2). X-ray diffraction (XRD) measurements revealed that pure Py3 films fabricated using the same method exhibited a strong diffraction peak centred at 11.5°, indicating its molecular stacking with long-range periodicity (Fig. 1a). By contrast, the film made by a SiO_x -molecule colloidal solution did not show any noticeable diffraction patterns, which implies that the presence of SiO_x NPs prevented the pure Py3 molecules from long-range periodic assembly on the substrate.

We utilized sub-1 cm⁻¹ high-resolution broadband sum-frequency generation vibrational spectroscopy (SFG-VS), which can selectively probe the vibrational spectra of interfacial molecules²⁷, to study the molecular packing at the interface (the experimental details are provided in Supplementary Note 1). As shown in Fig. 1b, the pure Py3 molecular film exhibited evident SFG-VS peaks at 3,055 cm⁻¹ and 3,025 cm⁻¹, both arising from the stretching vibration of aromatic C–H bonds, indicating the existence of a highly ordered self-assembled monolayer at the interface. In comparison, SFG-VS vibrational peaks were barely discernible in the SiO_x -molecule sample, suggesting a lack of orderliness in the interfacial molecular packing due to the involvement of SiO_x NPs (Supplementary Fig. 3 shows a detailed analysis of concentration dependence). Polarization-dependent Raman (p-Raman) spectroscopy

of the films (Fig. 1c) revealed that in a pure Py3 film, the intensity of the Raman peak at 1,609 cm⁻¹ stemming from the stretching of aromatic C=C bonds evidently varied with the polarized angle. This variation demonstrates the structural anisotropy, a salient characteristic of crystalline structures featuring directional molecular packing, which is consistent with the XRD results. Conversely, the intensity of the corresponding peak in the SiO_x -molecule film remained unchanged regardless of the polarized angle, implying a relative structural isotropy in the film lacking ordered molecular stacking. These highly aligned results prove SiO_x NPs to be able to interrupt the ordered molecular assembly at both bottom interface and upper layer. Atomic force microscopy (AFM) images of the film surface revealed a smooth surface morphology for the pure Py3 film, but a much larger surface roughness for the SiO_x -molecule film (Fig. 1d). This distinct surface feature underscores the presence of SiO_x NPs in the latter film to construct the bulk nano-heterointerface with the molecules. With the large contact area offered by SiO_x NPs, these molecules were no longer loosely bonded or stacked through to a single planar contact or weak van der Waals interaction. Instead, they can be securely fastened by the substantially increased surface area (Fig. 1e).

Formation mechanisms

We used ¹H nuclear magnetic resonance (NMR) spectroscopy to investigate the assembly of Py3 molecules in the SiO_x NP dispersion. As shown in Fig. 2a, we compared the NMR signals from pure Py3 molecules and Py3 with SiO_x NPs. The ¹H NMR signals arising from pyrene spanned over 7.90–8.40 ppm, and the detailed peak assignments are displayed in the inserted molecular structure of Py3 (the two-dimensional NMR results in Supplementary Fig. 4 show the peak assignments). On incorporating SiO_x NPs, all the peaks associated with the pyrene unit in Py3 underwent upfield shifts. These shifts can be attributed to the altered intermolecular stacking configuration of Py3, as a pyrene conjugation unit can impose a shielding effect to its neighbouring aromatic hydrogen atoms. The significant change in the molecular stacking configuration can also serve as an indication of the strong interaction of Py3 with SiO_x NPs. The comparison of the ³¹P NMR spectra of Py3 with and without SiO_x NPs revealed a peak shift in the ³¹P resonance of Py3 from 24.81 to 24.28 ppm on introducing SiO_x NPs (Fig. 2b), which further confirms molecular anchoring through the phosphonic group on the NPs.

To gain mechanistic insights into the assembly process in the dispersion, we used time-resolved (TR) dynamic light scattering (DLS) measurements to monitor the evolution of the colloidal size on mixing Py3 with SiO_x NPs (Fig. 2c). The dispersion of pure SiO_x NPs exhibited a sharp peak, indicative of a colloidal size of ~80 nm with a narrow size distribution (Supplementary Fig. 5), which is consistent with the transmission electron microscopy results (Supplementary Fig. 6). Initially, the pure Py3 solution hardly exhibited well-defined scattering peaks on stirring well, indicating that Py3 failed to form stable colloidal structures. However, on introducing SiO_x NPs to trigger the reaction, a broad peak at ~400 nm arose, pointing to the emergence of a distinct colloidal structure. This could be attributed to the presence of SiO_x NPs that induced the micellization of Py3, the colloidal size of which shrank over time as evidenced by the peak shift towards smaller sizes. This indicates that the micelles became more compact as the molecular assembly progressed. Meanwhile, the peak pertaining to the SiO_x NPs gradually shifted towards larger sizes accompanied by broadening, indicating the assembly of Py3 onto the surface of SiO_x NPs. As time progressed, these two distinct peaks gradually converged into a single peak, which stabilized after around 60 s, signifying the completion of the reaction and the finalization of molecular assembly. The stabilized colloidal size of SiO_x -Py3 was determined to be ~200 nm, exceeding the size of pure SiO_x NPs. This implies that the Py3 molecules have successfully bonded to the surface of SiO_x NPs. The zeta potential of the pure SiO_x NP dispersion was determined to be approximately -77 mV,

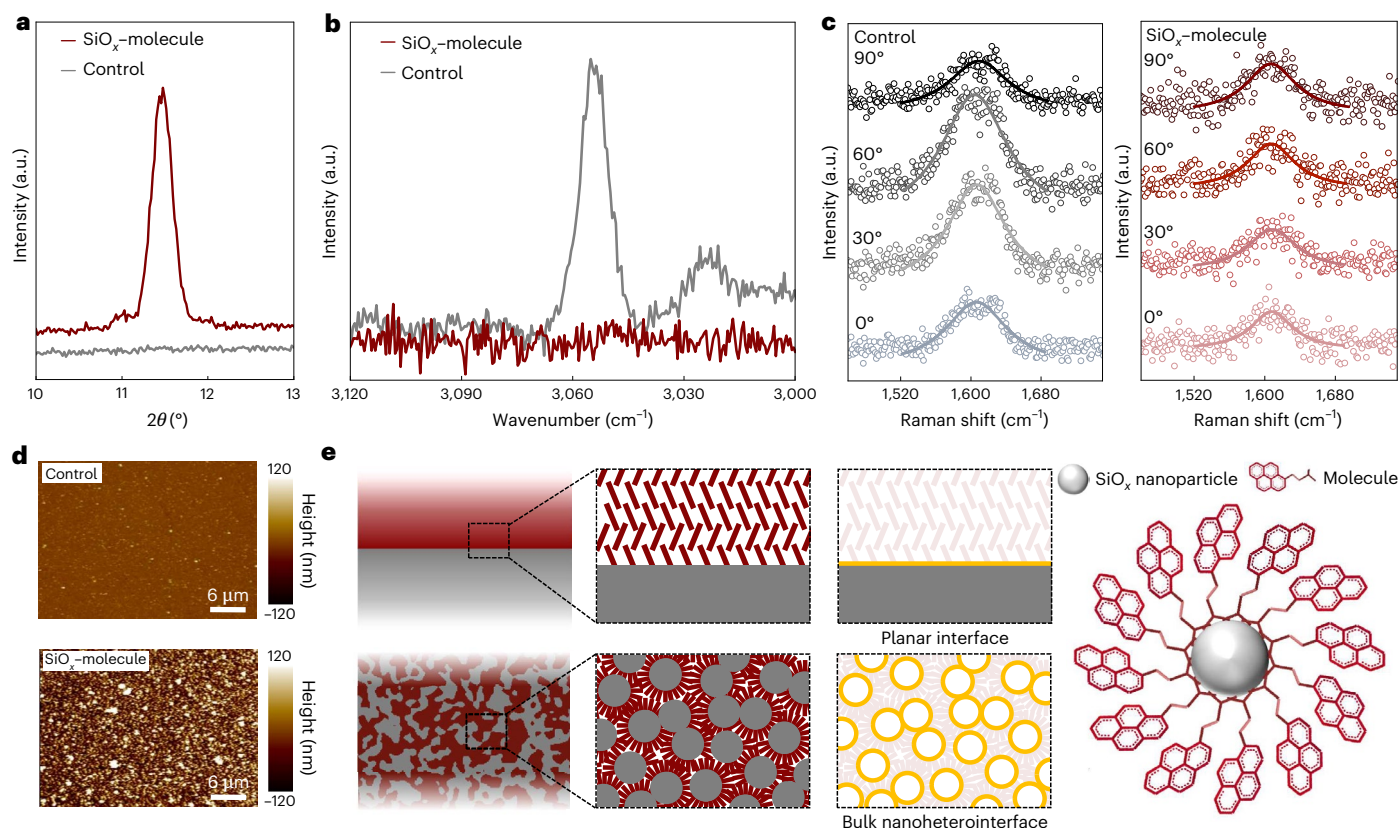


Fig. 1 | Construction of a bulk nano-heterointerface for molecule-based selective contacts. **a**, XRD patterns of the films made from pure molecules and SiO_x-molecule micelles. **b**, SFG-VS spectra of the films made from pure molecules and SiO_x-molecule micelles. **c**, p-Raman spectra of the films made from pure

molecules and SiO_x-molecule micelles collected under different polarization angles. **d**, AFM images showing the surface morphologies of films made from pure molecules (top) and SiO_x-molecule micelles (bottom). **e**, Schematic of the bulk nano-heterointerface compared with the conventional planar interface.

whereas SiO_x-Py3 showed a lower potential of −86 mV, indicating enhanced colloidal stability of SiO_x-Py3 (Supplementary Fig. 7).

We performed theoretical analysis on the interaction between SiO_x and Py3 using density functional theory. The result verifies that Py3 can strongly bind to the surface SiO_x through the phosphoric acid group (Fig. 2d). As reported elsewhere, phosphoric acid can readily react with the SiO_x surface through the elimination of a water molecule, resulting in the formation of a robust covalent bond^{28–33}. The calculated binding energy of −10.89 eV signifies a much stronger bond compared with the one between ITO and Py3 (−5.14 eV; Supplementary Note 2 shows the calculation details). To further confirm the stronger binding of Py3 with SiO_x than the ITO surface, we conducted Fourier transform infrared (FTIR) analysis on the samples of Py3 mixed with InO_x and SiO_x NPs. Given that the ratio of Sn in ITO is relatively small and surface interactions are primarily mediated by In sites, InO_x NPs were chosen as a representative model to mimic the ITO surface. The FTIR spectra of pure SiO_x and InO_x NPs were collected as controls. As shown in Fig. 2e, the FTIR spectrum of Py3 mixed with InO_x NPs closely resembled that of pure Py3, indicative of a minimal influence of InO_x NPs on the bond vibration in Py3. Specifically, the vibrational peaks associated with the P–O bonds, centered at 1,065 and 957 cm^{−1}, respectively, showed a negligible shift with the involvement of InO_x NPs, verifying the rather weak interaction of the phosphoric acid group with the InO_x surface. By contrast, the FTIR spectrum of Py3 changed significantly on mixing with SiO_x NPs. The vibrational peaks corresponding to the P–O bonds nearly vanished in the presence of SiO_x NPs, verifying the strong anchoring of the phosphoric acid group onto the SiO_x surface.

Due to the superior affinity of Py3 towards SiO_x compared with the ITO surface, on deposition on the ITO substrate, the ITO surface is

unable to rival the SiO_x NPs in binding with Py3. As a result, the strong contacts between Py3 molecules and SiO_x NPs persisted, ensuring the integrity of the colloidal structure of SiO_x-Py3 when processed into films on the ITO substrate. We utilized X-ray photoelectron spectroscopy (XPS) to verify the interplay between Py3 and SiO_x when the as-formed SiO_x-Py3 colloidal solution was processed into films. The XPS peak of P2p in the pure Py3 films was located at 134.06 eV, which shifted to a lower binding energy of 133.76 eV in the case of SiO_x-Py3 (Fig. 2f). Meanwhile, the Si2p peak shifted from 103.42 eV for pure SiO_x NPs to 103.80 eV for SiO_x-Py3 (Fig. 2g). The peak shifts observed in the XPS spectra confirm the binding between Py3 and SiO_x NPs in the film. This further supports the conclusion derived from the XRD, p-Raman and SFG-VS results that a disordered bulk nano-heterointerface was successfully established, enabling the assembly of Py3 to transcend merely relying on the planar and limited interface offered by the ITO.

Stability and universality

We compared the stability of the control and SiO_x-molecule films under external stresses via conductive atomic force microscopy (c-AFM). The evolution of the surface current signals of the films before and after continuous heating at 65 °C is displayed in Fig. 3a. After elongated heating for 40 days, the control film exhibited a significant reduction in its current signal, whereas the SiO_x-molecule sample remained almost unchanged, highlighting its superior resistance to thermal stress (Supplementary Fig. 8 shows the statistical current evolution). To corroborate its stability under thermal stresses, we further elevated the testing temperature to 85 °C and systematically assessed the structures and properties of the SiO_x-molecule films, which also exhibited negligible degradation (Supplementary Note 3). Similarly, the SiO_x-molecule

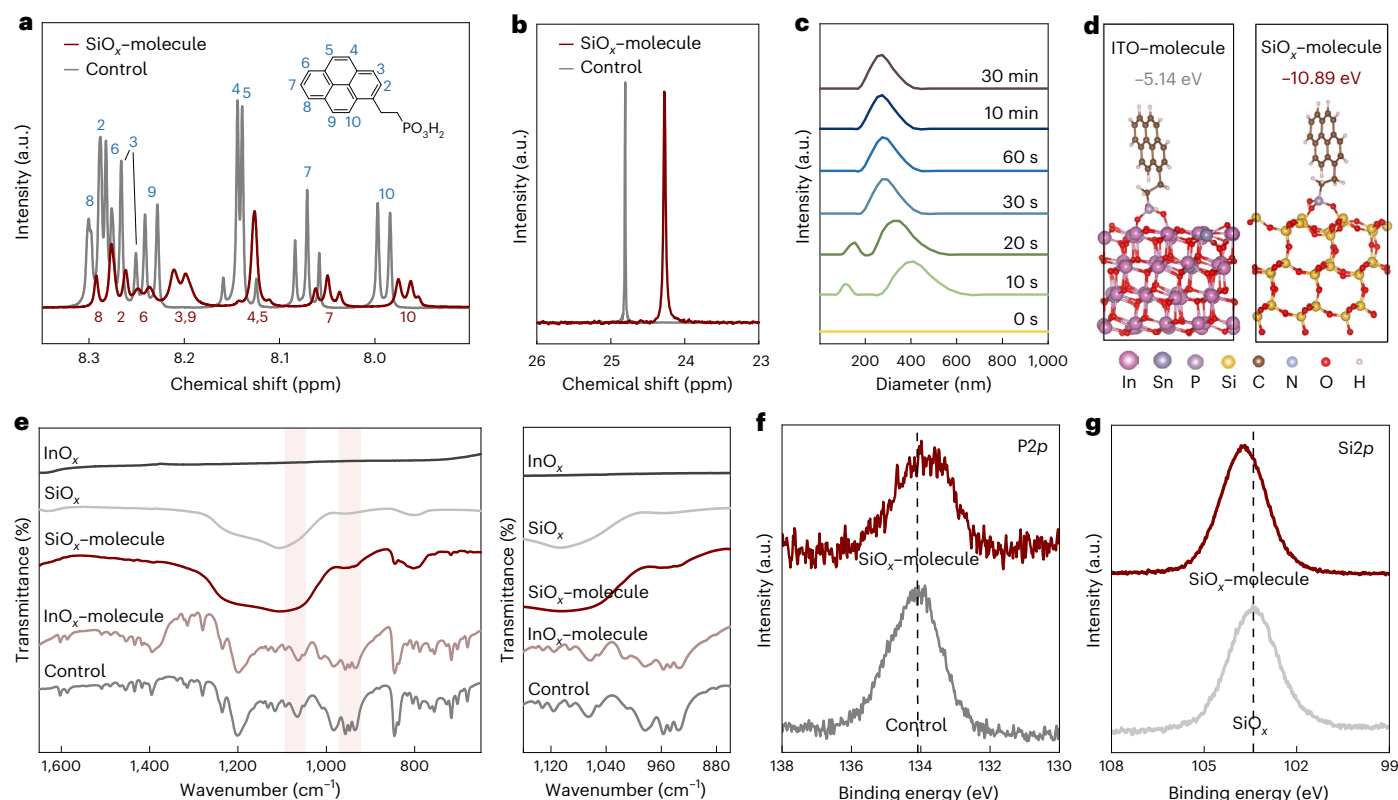


Fig. 2 | Formation mechanism of the nano-micelles and bulk nano-heterointerface. **a**, ^1H NMR spectra of molecules with SiO_x NPs compared with pure molecules. **b**, ^{31}P NMR spectra of molecules with SiO_x NPs compared with pure molecules. **c**, TR-DLS monitoring of the reaction between SiO_x NPs and molecules. **d**, Theoretical models of molecules bonded on the surface of ITO and SiO_x . **e**, FTIR spectra of pure molecules, pure SiO_x , pure InO_x , SiO_x molecule

and InO_x molecule. The red-shaded regions highlight the spectral windows of interest, which are enlarged in the right panel for clarity. **f,g**, XPS spectra of P2p (**f**) and Si2p (**g**) of the as-formed SiO_x -molecule film compared with the films of pure SiO_x and pure molecules. The dashed lines indicate the peak positions of the corresponding signals.

film exhibited greater resilience to mechanical stress, maintaining its initial current signal after 50 cycles of stretching, whereas the pure Py3 film exhibited a significant decline in the signal (Fig. 3b). The reduction in conductivity could be ascribed to the disrupted molecular stacking of Py3 with prolonged external stimuli. To verify this, we performed XRD measurements to monitor the structural evolution of the film during the heating process (Fig. 3c). The intensity of the diffraction peaks in the pure Py3 film underwent a gradual decline, accompanied by peak broadening, indicative of a reduction in crystallinity and disrupted molecular stacking. In comparison, throughout the aging period, the SiO_x -molecule film consistently exhibited no discernible diffraction peaks, implying that the film remained amorphous with the molecular assembly around the SiO_x NPs that lacked in long-range periodicity. We also evaluated the interfacial adhesion of the films by conducting tape peeling tests and mechanical bending tests. The results demonstrate the mechanical robustness and antidelamination of the SiO_x -molecule layer (Supplementary Note 3).

We further assessed the universality of this strategy by investigating two more typical molecules, namely, (2-(9H-carbazol-9-yl)ethyl) phosphonic acid (2PACz) and (2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl) phosphonic acid (MeO-2PACz), which have been in widespread use for charge-selective contacts. Similar to Py3, the XRD peaks of both 2PACz and MeO-2PACz films vanished on integrating SiO_x NPs (Fig. 3d). During the thermal test, the films with SiO_x NPs retained the initial amorphous phase (as revealed by the XRD results), whereas the pure molecular films showed decreased crystallinity. Similar phenomenon to Py3 was also observed in the SFG-VS results. As shown in Supplementary Fig. 9, pure molecular films of 2PACz and MeO-2PACz exhibited evident SFG-VS peaks, whereas the peaks were barely

discernible in the SiO_x -molecule samples. This verifies the lack of orderliness in the interfacial molecular packing due to the involvement of SiO_x NPs. The results of the p-Raman spectroscopy of the films also revealed structural anisotropy in the pure molecular films of 2PACz and MeO-2PACz, but a relative structural isotropy with the SiO_x NPs (Supplementary Fig. 10), which aligned well with the XRD and SFG-VS results. This strategy was also assessed for substrates with varying surface roughness (Supplementary Note 4 shows detailed results).

DLS measurements corroborated the formation of SiO_x -molecule micelles for both 2PACz and MeO-2PACz, manifesting as an increase in the colloidal size compared with the pure SiO_x NPs (Fig. 3e). Statistical analysis of the DLS results revealed an increase in the average colloidal size to ~200 nm on mixing SiO_x NPs with either 2PACz or MeO-2PACz (Supplementary Fig. 11). The XPS analysis on the films revealed that for the pure 2PACz and MeO-2PACz films, the peak of P2p —located at 134.12 and 134.00 eV, respectively—shifted to lower binding energies of 133.88 and 133.68 eV, respectively, when integrated with SiO_x NPs. The Si2p peaks for the SiO_x -2PACz and SiO_x -MeO-2PACz samples also showed shifts to higher binding energies of 104.06 and 103.60 eV, respectively, compared with pure SiO_x NPs, verifying the interaction between the phosphoric acid group and the SiO_x NPs (Supplementary Fig. 12). These findings strengthen the evidence of a reaction between these molecules and SiO_x NPs, resulting in the formation of a stable colloidal structure to securely fasten the molecules. The c-AFM monitoring of the surface current signals of the films during aging tests also demonstrates the superior stability of the films made from SiO_x NPs-molecule colloids. The average current signals of the films before and after thermal and mechanical aging tests are summarized in Fig. 3f. Although the current signals dramatically dropped for both pure 2PACz and MeO-2PACz

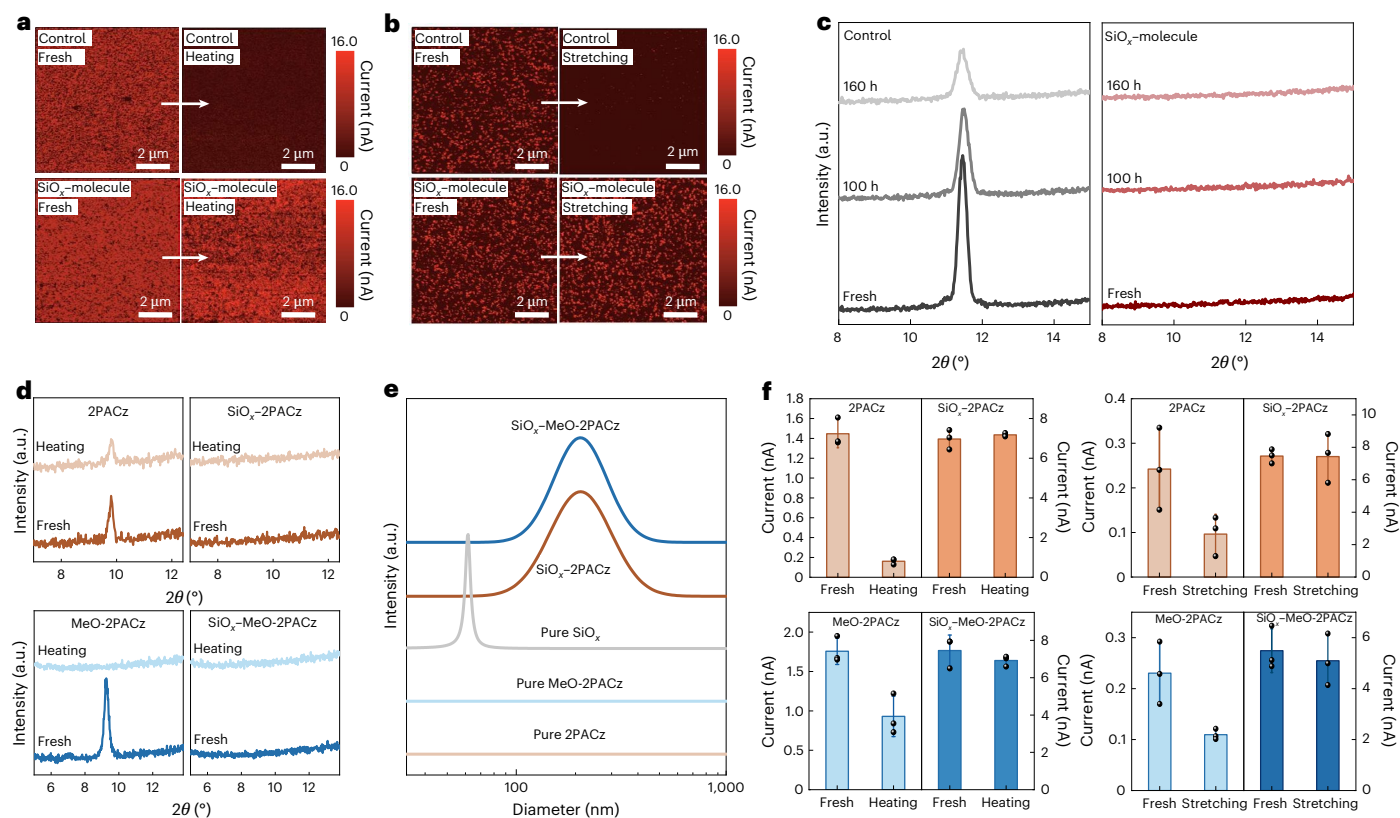


Fig. 3 | Stability of the films and the universality of the strategy. **a, b**, c-AFM images of control films and SiO_x -molecule films before and after **(a)** thermal stress and **(b)** mechanical stress. **c**, Evolution of the XRD patterns of the control films and SiO_x -molecule films during annealing. **d**, XRD patterns of the films of 2PACz, MeO-2PACz, SiO_x -2PACz and SiO_x -MeO-2PACz before and after heating. **e**, DLS results of the pure 2PACz solution, pure MeO-2PACz solution, pure

SiO_x NPs, SiO_x -2PACz and SiO_x -MeO-2PACz dispersions. **f**, Statistical surface current signals of the 2PACz, MeO-2PACz, SiO_x -2PACz and SiO_x -MeO-2PACz films before and after thermal and mechanical stresses. Data are presented as mean $\pm$ s.d. Sample size n is defined as the number of films measured by c-AFM: $n = 3$ for all cases.

films after heating and stretching (Supplementary Fig. 13 shows the c-AFM mapping results), the signals of the films made from the SiO_x NPs-molecule colloids showed negligible changes. These results prove the construction of the bulk nano-heterointerface as a universal strategy to enhance the stability of the molecular films for this purpose.

Photovoltaic performance

We evaluated the photovoltaic performances of the perovskite solar cells fabricated using SiO_x -molecule films as the hole-selective contacts, with the corresponding pure molecular films investigated in parallel for comparison (Supplementary Notes 5 and 6 show a detailed analysis of the device physics). The fabrication recipe for perovskite thin films was adapted from our previous report³⁴, with further optimizations made to the perovskite stoichiometry ($\text{FA}_{0.9}\text{Cs}_{0.05}\text{MA}_{0.05}\text{PbI}_3$). This composition is purely iodide based, possessing a more desirable bandgap for photovoltaics, and is readily adaptable to our scalable fabrication method. A champion PCE of 26.5% was achieved in the SiO_x -Py3-based device (a stabilized PCE of 26.4% and a certified stabilized PCE of 26.3%; Supplementary Figs. 14 and 15) for an aperture area of 0.1 cm^2 with an open-circuit voltage (V_{OC}) of 1.18 V, a short-circuit current density (J_{SC}) of 26.2 mA cm^{-2} and a fill factor (FF) of 85.6% (Fig. 4a). For the pure Py3-based device in the same batch, we obtained a PCE of 25.7% (stabilized, 25.3%; Supplementary Fig. 15) with a V_{OC} of 1.17 V, a J_{SC} of 26.0 mA cm^{-2} and an FF of 84.4%. According to the statistical results of PCEs obtained from 200 devices, the SiO_x -Py3-based PSCs exhibited a higher average value, verifying the superiority of SiO_x -Py3 in achieving a high PCE (Supplementary Fig. 16 and Supplementary Table 1 provide the detailed photovoltaic parameters). The increase in PCE

was also observed in 2PACz- and MeO-2PACz-based devices on using the SiO_x -molecule colloids for making the hole-selective layers (Supplementary Fig. 17).

We used high-resolution scanning transmission electron microscopy and energy-dispersive X-ray (EDX) spectroscopy to unveil the interfacial microstructure of the SiO_x -molecule-based device. Supplementary Fig. 18 displays the cross-sectional high-angle annular dark-field image obtained from scanning transmission electron microscopy, verifying a mesoporous-like interlayer crafted by SiO_x NPs. The corresponding EDX mapping of a more magnified area at the interface is presented in Fig. 4b, verifying the formation of an interpenetrating network between the SiO_x -molecule and perovskite (Supplementary Note 7 shows the detailed results and discussions).

To understand PCE improvements, we conducted a comprehensive analysis of the device physics to unveil the underlying mechanisms of the enhancements in photovoltaic parameters^{35,36}.

The nanoscale interfacial roughness, offered by the SiO_x -molecule colloidal structure, also rendered it easier for the perovskite precursor solution to spread out, as revealed by the contact angle measurements (Fig. 4c). When the perovskite precursor solution was applied onto the ITO substrate with the molecular films, the contact angle was determined to be 20.0° and 10.8° for Py3 and SiO_x -Py3, respectively. The decrease in contact angle, indicative of reduced interfacial tension, was also observed for 2PACz and MeO-2PACz when integrated with SiO_x NPs (Supplementary Fig. 19). The improved wettability would benefit the scalable fabrication method of perovskite devices (Supplementary Note 8 provides detailed discussions). We demonstrated its superiority in scalable fabrication by testing it in the blade-coating scenario of

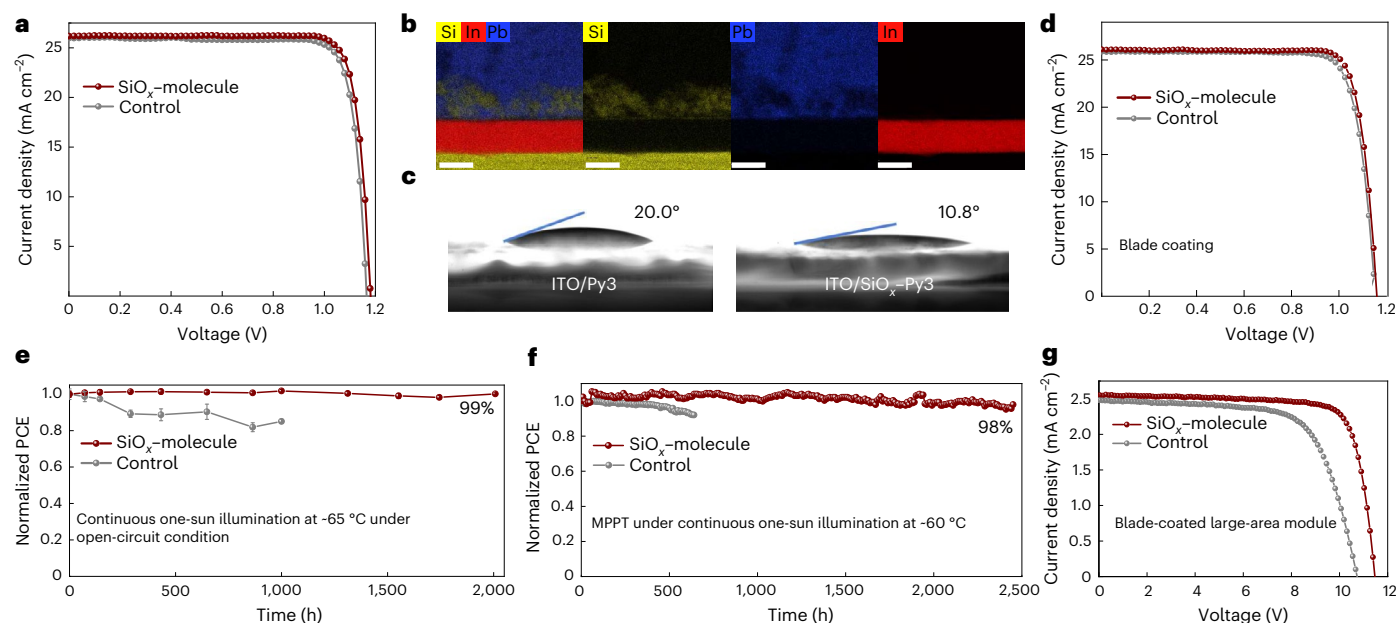


Fig. 4 | Photovoltaic performances. **a, j**–**V** curves of the perovskite solar cells fabricated using pure molecule and SiO_x–molecule films. **b**, Cross-sectional EDX mapping of the buried interface in the perovskite device fabricated using SiO_x–molecule colloids. Scale bar, 100 nm. **c**, Contact angles of the perovskite precursor solution deposited on the surface of the pure molecule and SiO_x–molecule films. **d**, **j**–**V** curves of the blade-coated perovskite solar cells. **e**, Evolution of the PCEs tracked under the open-circuit condition for continuous

one-sun illumination without a UV filter and thermal stress of -65 °C following the ISOS-L-2 protocol. Data are presented as mean $\pm$ s.d. Sample size n is defined as the number of devices tested: $n = 3$ for all cases. **f**, MPPT of the perovskite solar cells under continuous one-sun illumination without a UV filter at -60 °C for pure molecule and SiO_x–molecule films. **g**, **j**–**V** curves of blade-coated perovskite solar modules fabricated using the pure molecular film and SiO_x–molecule film.

perovskite devices. The small-area device made using the blade-coating method showed a high PCE of 25.1% for SiO_x–Py3, which was higher than that in pure-Py3-based device, proving the superiority of this strategy in terms of scalable fabrication (Fig. 4d; Supplementary Table 2 lists the photovoltaic parameters).

The stability of the perovskite devices was assessed under accelerated aging conditions according to the International Summit on Organic Photovoltaic Stability (ISOS) protocols³⁷. Under continuous one-sun illumination without an ultraviolet (UV) filter at -65 °C (ISOS-L-2 protocol), the PCEs of the devices based on SiO_x–Py3 retained over 99% after ~2,000 h, compared with over 20% PCE drop in the Py3-based device (Fig. 4e). We also measured the devices in MPPT under simulated one-sun illumination without a UV filter at -60 °C. The control device retained only ~90% of its initial PCE after 620 h of operation, whereas the SiO_x–Py3-based device retained over 98% of its initial PCE after 2,450 h (Fig. 4f; Supplementary Fig. 20 shows the MPPT certification).

The superior operational stability could be attributed to both the structural robustness of the bulk nano-heterostructure and the inherent stability of SiO_x compared with other oxides (Supplementary Note 9 provides detailed discussions on the comparison of other oxides). We performed comprehensive characterizations of the buried interface to verify the structural robustness of the bulk nano-heterostructure after device operation (Supplementary Note 10 shows detailed results and discussions). We subsequently extended this strategy to upscaled blade-coated solar modules with an aperture area of 32.5 cm² (Fig. 4g). The solar module based on SiO_x–Py3 achieved a champion aperture PCE of 23.0%, with a V_{oc} of 11.6 V, aJ_{sc} of 2.55 mA cm⁻² and an FF of 77.8%. By contrast, the best control device based on pure Py3 showed a PCE of only 18.1%, with a V_{oc} of 10.7 V, aJ_{sc} of 2.48 mA cm⁻² and an FF of 68.1%. This substantial improvement, as further verified by the statistical PCE results (Supplementary Fig. 21), underscores the particular superiority of our strategy for upscaled solar modules fabricated through the scalable blade-coating method.

Outlook

We report a bulk nano-heterointerface that reconstructed the assembly architecture of a molecule-based hole-selective layer as a more robust and universal alternative to the conventional molecular contact configuration that has been in widespread use but with stability issues in perovskite solar cells. The introduction of nano-colloids offered substantially increased contact area and enhanced binding forces for the molecules, rendering the molecular hole-selective layer with improved resistance to various external stimuli. This general interfacial architecture design reconciled the charge transport efficiency with structural durability, a long-standing issue limiting the longevity of perovskite devices. We believe this universal strategy for reconstructing the molecule-based selective contacts would unlock additional opportunities in architectural design to enhance molecular interfaces in various organic and organic–inorganic hybrid electronics, and encourage studies beyond the convention of planar contacts.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41563-026-02546-1>.

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Methods

Materials

All materials used in this work were obtained commercially and used without further purification, unless otherwise specified. The transparent ITO glass (10 Ω per square; transmittance, 88%) was purchased from Shenzhen Huayu United Technology. Transparent fluorine tin oxide glass (10 Ω per square) was purchased from Advanced Election Technology. The perovskite raw materials including lead iodide (PbI_2 ; purity, 99.999%; perovskite grade), methylammonium iodide (purity, 99.9%), formamidinium iodide (purity, 99.99%), methylammonium chloride (MACl; purity, 99.9%), lithium fluoride (LiF ; purity, 99.99%), C_{60} (purity, 99.9%) and SiO_x dispersions (with a hydrated diameter of 100 nm, dispersed in ethanol) were purchased from Advanced Election Technology. InO_x (diameter, 30 nm; purity, 99.99%) was purchased from LiaoNing JiaYun High Technology. Bathocuproine (BCP; purity, 99%), caesium iodide (purity, 99.999%), propylammonium chloride (purity, 99.8%) and oleylammonium iodide (purity, 99.5%) were purchased from Xi'an Yuri Solar. L- α -phosphatidylcholine (purity, 99%) was obtained from Shanghai Macklin Biochemical Technology. 2PACz (purity, 98%) and MeO-2PACz (purity, 98%) was obtained from Tokyo Chemical Industry. The required solvent including *N,N*-dimethylformamide (DMF; extra dry; purity, 99.8%), dimethyl sulfoxide (anhydrous, $\geq 99.9\%$), 2-methoxyethanol (2-ME; extra dry; purity, 99.9%), *N*-methyl-2-pyrrolidone (extra dry; purity, 99.9%), isopropanol (IPA; extra dry; purity, 99.5%), methanol (extra dry; purity, 99.8%), anisole (extra dry; purity, 99.5%) and silver (Ag; purity, 99.99%) were purchased from Sigma-Aldrich.

Preparation of molecule solutions, SiO_x -molecule colloids and their bulk films

For the preparation of molecule solutions, the molecule powders were fully dissolved into a mixed solvent of ethanol and DMF at a volume ratio of 3:1 at room temperature. For the preparation of SiO_x -molecule colloids, 10, 25, 40, 50 or 100 μl of SiO_x dispersions was mixed with 1 ml of 3 M pure molecule solution to initiate the reaction. The mixture was then stirred for 30 min at room temperature in a nitrogen glovebox to ensure the complete reaction between SiO_x NPs and the molecules. To prepare the films, filtered Py3, 2PACz and MeO-2PACz solutions or SiO_x -Py3, SiO_x -2PACz and SiO_x -MeO-2PACz dispersions were spin coated at 1,000 r.p.m. for 10 s and 3,000 r.p.m. for 40 s in the nitrogen glovebox. Afterwards, the films were annealed at 115 $^\circ\text{C}$ for 20 min. For the comparison test of a sequential deposition, SiO_x NPs and molecule solutions were sequentially spin coated on the ITO substrate, followed by annealing at 115 $^\circ\text{C}$ for 20 min.

Preparation of perovskite precursors and films

For the preparation of perovskite films with the composition of $\text{FA}_{0.9}\text{MA}_{0.05}\text{Cs}_{0.05}\text{PbI}_3$, 1.65 M of the perovskite precursor solution was prepared by dissolving caesium iodide, formamidinium iodide, methylammonium iodide and PbI_2 in 1 ml of mixed solvent of DMF and dimethyl sulfoxide at a volume ratio of 4:1. Additional 7 mol% of PbI_2 and 12 mol% of MACl was added to the precursor solution, and the solution was stirred overnight. The filtered perovskite precursor was then spin coated on a substrate at 1,000 r.p.m. for 10 s and 6,000 r.p.m. for 40 s in the nitrogen glovebox. Then, 200 μl of anisole antisolvent was dropped quickly at 25 s during the second spinning step. Afterwards, the film was annealed at 115 $^\circ\text{C}$ for 30 min.

Fabrication of inverted perovskite solar cell devices

The prepatterned ITO glasses (20 mm $\times$ 20 mm) were washed with detergent solution, deionized water, acetone and IPA, each under 30 min of sonication, followed by drying with a pure nitrogen flow. The cleaned and dried ITO substrates were then treated with UV-ozone for 30 min before transferring into glovebox filled with nitrogen. The molecule solution or the colloid was spin coated (Jiangsu Lebo

Science Instrument) at 1,000 r.p.m. for 10 s and 3,000 r.p.m. for 40 s in the nitrogen glovebox (to obtain the optimal PCE of the device, the amount of SiO_x dispersions was 25 μl for the preparation of SiO_x -Py3 colloids). Afterwards, the film was annealed at 115 $^\circ\text{C}$ for 20 min. After cooling down to room temperature, the perovskite films were coated on these layers following the abovementioned method. For the surface passivation layer, oleylammonium iodide was dissolved in IPA with a concentration of 0.5 mg ml^{-1} and stirred at room temperature for 2 h. The oleylammonium iodide solution was spin coated on top of the as-prepared perovskite at 5,000 r.p.m. for 30 s, and then transferred to the hotplate and annealed at 100 $^\circ\text{C}$ for 5 min.

After cooling down to room temperature, the whole device was transferred to a vacuum chamber under a base pressure of $<5.0 \times 10^{-6}$ Pa. Then, 1.0 nm of LiF at a rate of 0.1 $\text{\AA s}^{-1}$, 30 nm of C_{60} at a rate of 0.3 $\text{\AA s}^{-1}$ and 8 nm of BCP at a rate of 0.2 $\text{\AA s}^{-1}$ were thermally evaporated on the perovskite thin film sequentially (Angstrom Engineering and Beijing Technol Science), respectively, without breaking the vacuum. Finally, a 100-nm-thick Ag electrode at a rate of 1.5 $\text{\AA s}^{-1}$ was deposited to complete the devices for current density-voltage (J - V) measurements or stability tests.

Fabrication of blade-coated perovskite solar cell devices

The prepatterned ITO glass substrates were cleaned by sequentially washing with detergent, deionized water, acetone and IPA. Before use, ITO was cleaned with UV-ozone for 20 min. Then, the substrate was spin coated with a thin layer of NiO_x NP film (20 mg ml^{-1} of NiO_x water dispersion) at 1,500 r.p.m. for 30 s, and annealed in ambient air at 125 $^\circ\text{C}$ for 10 min. Thereafter, 0.5 mg ml^{-1} of Py3 solution or SiO_x -Py3 dispersion was deposited on NiO_x at 3,000 r.p.m. for 40 s. Then, 1.4-M $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ was prepared by directly mixing FAPbI_3 (in 2-ME) and CsPbI_3 (in 2-ME) diluting with a mixed solvent of 2-ME and *N*-methyl-2-pyrrolidone at a volume ratio of 10:1 before blade coating. Also, 30 mol% of MACl, 10 mol% of propylammonium chloride and 0.2 mg ml^{-1} of L- α -phosphatidylcholine were added into the precursor solutions as additives. Subsequently, the precursor solution was blade coated onto the substrates with a gap of 120 μm at a coating speed of 15 mm s^{-1} . The relative humidity was 30%. A N_2 air knife was used to facilitate drying throughout the coating process. The wet perovskite films were then transferred onto a hotplate and annealed at 150 $^\circ\text{C}$ for 5 min in air. The solar cells were completed by thermally evaporating 1 nm of LiF , 30 nm of C_{60} , 6 nm of BCP and 150 nm of Ag.

Fabrication of blade-coated perovskite solar modules

For the preparation of an FA-based $\text{FA}_{0.95}\text{Cs}_{0.05}\text{PbI}_3$ thin film, the perovskite precursor solution was prepared by mixing 1.5 M of PbI_2 , 1.33 M of formamidinium iodide, 0.07 M of caesium iodide, 0.12 M of propylammonium chloride and 0.3 M of MACl in a mixed solvent of 2-Me and *N*-methyl-2-pyrrolidone at a volume ratio of 10:1 with additions of 0.05 mg ml^{-1} of PEACl, 0.2 mg ml^{-1} of L- α -phosphatidylcholine and 2 mg ml^{-1} of RbI. The solution was stirred for 1 h. The filtered perovskite precursor solution was then blade coated on a substrate at 15 mm s^{-1} with a blade gap of 150 μm under dry-air quenching. The films were annealed at 140 $^\circ\text{C}$ for 5 min. The blade-coating processes were performed under the ambient condition with controlled humidity ($\sim 30\%$). After cooling down to room temperature, the whole device was transferred to a vacuum chamber under a base pressure of $<5.0 \times 10^{-6}$ Pa. Then, 1.0 nm of LiF at a rate of 0.1 $\text{\AA s}^{-1}$, 30 nm of C_{60} at a rate of 0.3 $\text{\AA s}^{-1}$ and 8 nm of BCP at a rate of 0.2 $\text{\AA s}^{-1}$ were thermally evaporated on the perovskite thin film sequentially (Angstrom Engineering and Beijing Technol Science), respectively, without breaking the vacuum. Finally, 100-nm-thick Ag electrode at a rate of 1.5 $\text{\AA s}^{-1}$ was deposited. The perovskite solar modules consisted of ten perovskite subcells connected in series. Between each subcell, P1, P2 and P3 lines were laser scribed (HANSPV PV-HL0303) to form the typical series connection of perovskite solar modules. The minimodules were composed of ten

subcells, each with a width of 6.0 mm and a dead-area width of 100 μm (Supplementary Fig. 22 shows the optical image), leading to a geometry filling factor of 98.3%.

Device characterizations

J - V characteristics of photovoltaic cells were taken using a Keithley 2450 source measure unit under a simulated AM1.5G spectrum, with an SS-X100R solar simulator (Enlitech). The illumination light was calibrated to 100 mW cm^{-2} of AM1.5G using a silicon reference cell. Typically, the devices were measured in the reverse scan (1.25 V $\rightarrow$ 0 V, step of 0.02 V). All the devices were measured with an aperture area of 0.1 cm^2 . The measurements were done without preconditioning such as light soaking and the bias voltage was applied. The steady-state PCE was calculated by measuring a stabilized photocurrent density under a constant bias voltage. External quantum efficiencies were measured using an integrated system (Enlitech) and a lock-in amplifier with a current preamplifier under the short-circuit condition.

The electrochemical impedance spectroscopy measurement was performed on a ZENNIUM PRO electrochemical workstation from 106 to 10 Hz with a disturbance a.c. signal of 10 mV (d.c. bias, 0.5 V).

Transient photocurrent decay under the short-circuit condition was analysed using a photoelectrochemical measurement system. The transient photovoltage decay under the open-circuit condition was conducted using the same photoelectrochemical measurement system (CIMPS-abs + UBSC-LD50XH).

Thermal admittance spectroscopy was conducted on samples prepared following the device fabrication conditions above. The test was done using an Agilent 4990A precision LCR meter. The scanning range of the a.c. frequency (f) was 0.02–2,000 kHz. The d.c. bias (V) was fixed at 0 V.

The drive-level capacitance profiling measurements were performed using a Keysight E4980A precision LCR meter. Here the d.c. bias was scanned from 0 V to V_{oc} (for example, 1.1 V) for the perovskite solar cell. Furthermore, this method uses a series of variable δV (for example, 20–200 mV) to measure the junction capacitance and acquire the capacitance contribution from the trap states by taking advantage of the information embedded in the higher-order terms. The capacitance measured at each δV was recorded and fitted with a polynomial function to obtain C_0 and C_1 .

$$C = C_0 + C_1\delta V + C_2(\delta V)^2 + \dots$$

With the determination of C_0 and C_1 , the carrier density (N) that includes both free carrier density (N_0) and trap density (N_T) at a certain position X from the junction barrier is calculated as

$$N = -\frac{C_0^3}{2q\epsilon A^2 C_1},$$

where q is the elementary charge, ϵ is the dielectric constant of the semiconductor and A is the active area of the junction. The profiling distance from the junction barrier was calculated as $\epsilon A/C_0$. For the comparison of trap densities, the trap densities measured at 10 kHz are representatively given in this work, which would not affect the comparison of the relative values as the trap densities are basically on the same order of magnitude with the change in a.c. frequency in the vicinity of 10 kHz.

Cross-sectional Kelvin force probe microscopy (KPFM) was performed using a commercial AFM NX-10 system (Park Systems) housed in a N_2 -filled glovebox (MBrun). The device was mechanically cleaved to expose the cross-section, eliminating the need for ion-beam milling and thereby preventing beam-induced damage during sample preparation. A region with a root mean square roughness below 90 nm was chosen to reduce errors associated with surface irregularities. To accurately observe the contact potential difference of every sample,

the biased tip was calibrated at a scan rate of 0.25 Hz before and after each measurement using highly ordered pyrolytic graphite, known for its consistent work function of 4.65 eV. The Spark 70 Pt tip was adopted to maximize the image resolution of the work function by considering high electrical sensitivity. A custom-made device holder was used to mount the prepared sample, with the sample stage acting as the ground. The contact potential difference (CPD) profiles obtained under LED illumination at an intensity of 25 mW cm^{-2} , measured under both open- and short-circuit conditions, were used to calculate the electric field distribution along the device cross-section based on the following equation given by

$$E(z) = -\frac{d}{dz}V(z) = -\frac{d}{dz}\text{CPD}(z) = -\frac{d}{dz}\left[\frac{\phi(z)_{\text{sample}} - \phi_{\text{tip}}}{e}\right],$$

where $E(z)$ is the electric field difference, $\phi(z)_{\text{sample}}$ and ϕ_{tip} are the work function of the sample and probe, respectively, and e is the elementary charge. Since the electric potential is expressed in the form of Poisson's equation as

$$\rho = \epsilon\epsilon_r\nabla^2\text{CPD},$$

where ρ is the charge density and ϵ and ϵ_r are vacuum and relative permittivities, respectively, the corresponding accumulated charge density can be represented through the equation given by

$$\Delta\rho_{\text{AC}} = \rho_{\text{OC}} - \rho_{\text{SC}} = \epsilon\epsilon_r\nabla^2(\text{CPD}_{\text{OC}} - \text{CPD}_{\text{SC}}),$$

where ρ_{AC} is the accumulated charge density and ρ_{OC} and ρ_{SC} are the charge density under the open- and short-circuit conditions, respectively. The data were smoothened using Savitzky–Golay processing by a second-order polynomial regression and 20 data points of ratio.

Device stability tests

The stability test following the ISOS-L-2 protocol was conducted under continuous heating at -65°C , AM1.5G illumination (100 mW cm^{-2} , white LED) and open-circuit condition. The PCEs of the devices were tracked over time, and the J - V measurements were performed after cooling the devices down to room temperature. The MPPT of the devices was conducted under AM1.5G illumination at -60°C . The voltage at the maximum power point was automatically adjusted and applied, and the power output of the devices was tracked over time. For the temperature control, we used the PR-SCCS-C24Q Solar Cell Stability Test System (PURI Materials), in which eight devices can be tracked and the devices were placed on a heating stage, with a temperature detector attached and connected to an external proportional–integral–derivative controller. For all the stability measurements, the devices were tested without any pretreatment or aging. There is a UV component in the light source and no UV filter is applied for the stability tests.

Materials characterization

The Jupiter XR Oxford Instruments were used for the c-AFM measurements of the molecular films with an ASYLEC 01-R2 tip ($f = 75$ kHz, $k = 2.8$ N m^{-1} , tip coating of Ti/Ir), and the c-AFM results were obtained with a setpoint of 0.2 V under a bias of 400 mV, unless stated otherwise. A bias of -20 mV was applied for the measurements of ITO and ITO/ SiO_x , where the current flow was allowed along the ITO surface without signal saturation. For Py3 and SiO_x -Py3 films, the SiO_x -molecule films were deposited on a rigid ITO glass substrate for the thermal stability test, the aged films were obtained by heating the fresh films in a nitrogen glovebox at 65°C for 40 days during which the AFM and c-AFM results were collected at 10, 20 and 40 days. For the 2PACz-, SiO_x -2PACz-, MeO-2PACz- and SiO_x -MeO-2PACz-based films, the measurements were done before and after heating the films at 65°C for

10 days. For the tests of the resistance of films on mechanical stress, the SiO_x -molecule films were deposited on a flexible PET substrate (Supplementary Fig. 23 shows the surface morphologies of the two different substrates measured by c-AFM). The preparation of the precursor solution and fabrication procedures are the same as the aforementioned method. The films were stretched with 3% engineering strain for 50 cycles before the measurements. Jupiter XR (Oxford Instruments) was used for the KPFM measurements for assessing the stability of the molecular layers in realistic device fabrication scenarios, including the effects of solvent rinsing and perovskite crystallization. Cypher VRS (Oxford Instruments) was used for the KPFM measurements of the molecular films with an ASYLEC 01-R2 tip ($f = 75 \text{ kHz}$, $k = 2.8 \text{ N m}^{-1}$, tip coating of Ti/Ir). The tip-sample distance (delta height) was set to be 50 nm. To evaluate the interfacial adhesion of the SiO_x -Py3 layer, we performed mechanical bending tests. For these tests, the SiO_x -Py3 layer was deposited onto a flexible ITO substrate followed by the deposition of the perovskite and the other functional layers to attain a full device stack. After bending, we washed the top layers and performed the AFM measurements.

To probe the carrier transport behaviour of the perovskite at the bulk heterointerface, we conducted c-AFM measurements on the delaminated perovskite, with the buried interface being exposed. The delaminated perovskite films were attached onto an insulating substrate, which is the insulating polymer used for perovskite layer peel-off. A conductive tape is attached to the perovskite surface, with the other end connected to the c-AFM equipment. This setup allows the current to flow from the c-AFM tip laterally along the perovskite surface before being collected, which can minimize the bulk effect and better reflect the intrinsic properties of the interfacial perovskite part in contact with SiO_x -Py3.

Contact angle measurements were done with high precision ($\pm 0.1^\circ$ accuracy) using a CA200 contact angle analyser.

For the SFG-VS measurement, the sample films were prepared following the same procedure used for device fabrication. SFG-VS studies were carried out using the custom sub-centimetre high-resolution broadband SFG-VS system. The SFG-VS setup is made by means of the precise synchronization of two laser systems: a Ti:sapphire femtosecond laser system ($\sim 7 \text{ W}$, $\sim 35 \text{ fs}$, 800 nm , repetition rate of 1 kHz ; Astrella, Coherent) and a picosecond neodymium-doped yttrium aluminium garnet laser system (2 W , $\sim 80 \text{ ps}$, $1,064 \text{ nm}$, repetition rate of 1 kHz).

For the high-resolution XPS measurements, the samples were loaded into the XPS instrument (Thermo Fisher ESCALAB Xi+) and Al $K\alpha$ ($1,486.6 \text{ eV}$) X-rays were used as the excitation source.

For the ultraviolet photoelectron spectroscopy measurements, a He discharge lamp, emitting UV energy at 21.2 eV , was used for excitation. All ultraviolet photoelectron spectroscopy measurements were performed using standard procedures with a bias of $\sim 5 \text{ V}$ applied between the samples and detectors. Clean gold was used as a reference.

The high-angle annular dark-field images and EDX mapping of the cross-sectional devices were investigated using a field-emission scanning electron microscope (Talos F200X G2, Thermo Fisher Scientific) equipped with EDX and high-angle annular dark-field detectors in the scanning transmission electron microscopy mode at an operating voltage of 200 kV . For sample preparation, a cross-sectional lamella was created and the cross-sectional scanning electron microscopy was performed using a scanning electron microscope (Helios 5 UX, Thermo Fisher Scientific) with an EasyLift nanomanipulator, aided by a Ga-focused ion beam. Protective carbon and platinum layers were deposited on the sample using electron and ion beams to prevent damage.

The polarized Raman measurement was realized on an Alpha300RAS (WITec) Raman system. A diode-pumped solid-state laser (532 nm , Cobolt lasers) was focused on the sample with a diffraction-limited beam size of 900 nm by a long-working-distance $\times 50$ ZEISS LD EC Epiplan-Neofluar Dic objective (numerical aperture = 0.55).

A G2 grating (600 g mm^{-1} , BLZ = 500 nm) was used. To get a better spectral signal-to-noise ratio, the laser power was set to 15 mW and the integration time of collecting the spectrum was set to 5 s . An analyser was inserted to polarize the incident laser beam. The angle of the analyser was set as 0° , 15° , 30° and 45° . The polarization angle is twice the analyser angle, namely, 0° , 30° , 60° and 90° , respectively.

For the FTIR measurements, Py3 was dissolved in a mixed solvent of ethanol and DMF at a volume ratio of 3:1. Subsequently, SiO_x or InO_x NPs were added to the Py3 solution and stirred at room temperature for 30 min . The solutions were deposited onto glass substrates and dried at 115°C in a nitrogen glovebox to remove the solvent before the measurements. SiO_x -2PACz, SiO_x -MeO-2PACz, InO_x -2PACz and InO_x -MeO-2PACz samples were prepared using the same method. The FTIR spectra were collected at room temperature using an FTIR spectrometer (Thermo Fisher Nicolet iS50) equipped with a deuterated triglycine sulfate KBr detector. The spectral resolution was set to 1 cm^{-1} and the aperture was set to 4 mm , and the spectra were acquired by averaging 64 scans.

NMR experiments were performed at 25°C on a Bruker NEO 600-MHz NMR spectrometer (600.23 MHz for proton frequency) equipped with a QCI-F Cryoprobe. For one-dimensional ^1H experiments, 64,000 complex data points were acquired with 16 scans. Two-dimensional ^1H - ^1H NOESY data using the 'noesygpphpp' pulse sequence were collected with $2 \text{ scans} \times 4,096 \text{ data points (F2)} \times 512 \text{ increments (F1)}$ with mixing times of 50 ms and 700 ms , respectively. Two-dimensional ^1H - ^1H COSY data using the 'cosygppppqf' pulse sequence were recorded with $1 \text{ scan} \times 4,096 \text{ data points (F2)} \times 512 \text{ increments (F1)}$. The ^{31}P one-dimensional experiments were performed at 25°C on a Bruker NEO 500-MHz NMR spectrometer with an RT Cryoprobe.

For the XRD measurements, a 6-mM solution of the pure molecule was prepared by fully dissolving the molecule powder in a mixed solvent of ethanol and DMF at a volume ratio of 3:1 at room temperature. For the preparation of SiO_x -molecule colloids, $50 \mu\text{l}$ of SiO_x NP dispersion was added to 1 ml of the 6-mM molecular solution. Thick films of both pure molecule and SiO_x -molecule colloids were fabricated by spin coating at 500 r.p.m. for 20 s , followed by $3,000 \text{ r.p.m.}$ for 10 s in the nitrogen glovebox and then annealed at 115°C for 20 min . The signals were collected in the reflection mode at room temperature on a D8 Advance diffractometer equipped with a one-dimensional LynxEye detector using monochromated $\text{Cu K}\alpha$ ($\lambda = 1.5418 \text{ \AA}$) radiation. For the stability tests, the aged films were obtained by heating the fresh films at 110°C for 100 h and 160 h and cooling down to room temperature before the XRD measurements. The grazing-incidence X-ray diffraction measurements were performed on a Bruker D8 Discover XRD device equipped with high-brilliance rotational point-focused Cu X-ray source ($\lambda = 1.5418 \text{ \AA}$), equipped with DUO detectors of scintillation counter and LynxEye. The in situ XRD measurement for perovskite films was conducted at 298 K , performed on a Bruker D8 Venture diffractometer, equipped with a metal-jet microfocus source ($\text{Ga K}\alpha$ radiation, $\lambda = 1.34138 \text{ \AA}$), PHOTON III detector in the shutterless mode and Oxford 1000 Plus cryosystem. The in situ monitoring was done with an X-ray exposure of 0.5 s for each frame, and the integration of frames was processed using the APEX 4 software package.

For the PL measurements, the perovskite layers were deposited onto the molecular layers using the same method as the device fabrication conditions. To obtain signals from different depths, the laser beam was adjusted to be 600 nm or 380 nm when incident on the samples. For the time-resolved photoluminescence (TRPL) layers were deposited onto the molecular layers using the same method as the device fabrication conditions. The sample was excited using a picosecond pulsed diode laser (PicoQuant LDH 450), with an $\sim 70\text{-ps}$ pulse width and 20-MHz repetition rate, focused on the sample with a $\times 100$ objective (numerical aperture = 0.90). The photoluminescence (PL) signal

was acquired through the time-correlated single-photon-counting StrobeLock system. The total instrument response function for the PL decay was less than 200 ps, and the temporal resolution was less than 30 ps. The energy density of the laser for TRPL measurements was set to be 2.6 nJ cm^{-2} . For scanning electron microscopy, the surface and cross-sectional morphologies of the perovskite films were carried out on a QuattroS field-emission environment scanning electron microscope. For TRPL mapping, the samples were prepared following the device fabrication conditions. Two-photon luminescence microscopy and line-scan TRPL were measured using a PicoQuant MicroTime 100 confocal microscopic system. An Olympus $\times 100$ oil-immersion objective with a numerical aperture of 1.45 was used for the measurement. Samples were covered by immersion oil (Olympus Immersion Oil Type-F) in a dark environment during the measurement. A picosecond 640-nm pulsed laser with a repetition rate of 20 MHz was used to measure the cross-sectional PL mapping of $\text{FA}_{0.9}\text{MA}_{0.05}\text{Cs}_{0.05}\text{PbI}_3$ perovskite films. The average laser intensity was kept at around 36 nW at 20 MHz, corresponding to a photon energy of 1.8 fJ per pulse. For the line-scan TRPL measurement, a series of selected points on a line were excited by the 640-nm pulsed laser and the integration time for each point was 120 s. The repetition rate of the 640-nm laser was set at 0.5 MHz to ensure the complete decay of the fluorescence signal.

The DLS and zeta potential measurements were realized on a BI-200SM/NanoBrook ZetaPALS/BI-DNDC system, and a constant laser intensity was maintained once set. To perform time-resolved tests on the sample, the laser was stabilized and maintained at a constant laser intensity throughout the test process.

Tape-peeling tests were performed on an INSTRON 5944 testing system with a 5-N load cell, under a loading rate of 50 mm min^{-1} . Scanning electron microscope tests were performed using ZEISS Gemini460. Infrared thermal imaging was performed using a Hik-microtech HM-TPH21Pro-3AQF Handheld Thermometer Thermal Imager. The thermal diffusivity was measured by a Netzsch LFA 457 instrument. Scanning thermal microscopy was performed using Oxford Cypher VRS.

The time-of-flight secondary ion mass spectrometry measurements (ULVAC-PHI NanoTOF 3, with a double microchannel plate detector and triple-ion-focusing time-of-flight analyser) were performed using pulsed primary ions from an Ar (2 kV) gun for sputtering and a Bi^+ pulsed primary ion beam for analysis (30 kV). The area of analysis was $100 \times 100 \text{ }\mu\text{m}^2$, whereas the sputtering area was $400 \times 400 \text{ }\mu\text{m}^2$.

The PL quantum yield measurements of perovskite films were collected by a fibre-coupled spectrometer, paired with a 405-nm power-tunable continuous-wave laser to photoexcite the films within an integrating sphere. The power of the excitation light is precisely measured utilizing a calibrated power meter. The test was performed using steady-state and time-resolved photoluminescence spectrometers (FLS1000 + FS5 + Pico1000).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data needed to evaluate the conclusions in the paper are present in the article or its Supplementary Information. Source data are provided with this paper.

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Author contributions

J. Xue conceived the idea and supervised the project. Y.L. and J.S. performed the experiments and data analysis under the supervision of J. Xue. K.Z., S.C. and L.J. fabricated the solar cell devices. B.-j.Z. and L.Z. performed the SFG-VS measurements under the supervision of H.-f.W. C.D. and I.Y. conducted the theoretical calculations. Q. Liu synthesized the molecules. Y.T., X.S., L.Y., X.M., Q. Li, R. Li, H.X., J. Xu, J.Z. and D.J. assisted with the characterizations and device fabrication. S.-G.C. performed the cross-sectional KPFM under the supervision of J.-W.L. H.Z. performed the cross-sectional TRPL mapping under the supervision of Z.N. R. Liu, R.W., H.J.Y. and D.Y. provided helpful discussions. J. Xue wrote the paper. All authors discussed the results and commented on the paper.

Competing interests

The authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes

0.1 cm²

☐ No

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes

the aperture area of the device is 0.1 cm², designated by the shadow mask

☐ No

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☐ Yes

We have provided the steady-state output data in Supplementary Fig. S17

☒ No

Voltage scan conditions

☒ Yes

reverse scan (1.25 V → 0 V, step 0.02 V).

☐ No

Explain why this information is not reported/not relevant.

Test environment

☒ Yes

25 °C, N₂ gloveboxes

☐ No

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☐ Yes

Provide a description of the protocol.

☒ No

All the devices were measured without pre-conditioning such as light-soaking and applied a bias voltage

Stability of the J-V characteristic

☒ Yes

Supplementary Fig. S17

☐ No

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☐ Yes

Provide a description of hysteresis or any other unusual behaviour observed during the characterization.

☒ No

We have provided the steady-state output data in Supplementary Fig. S17

Related experimental data

☒ Yes

The steady-state output data is provided in Supplementary Fig. S17

☐ No

Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes

External quantum efficiencies (EQEs) were measured using an integrated system (Enlitech) and a lock-in amplifier with a current preamplifier under short-circuits' condition.

☐ No

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator

☒ Yes

Supplementary Fig. S122

☐ No

Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	Provide a description of the measurement conditions. This paper focused on single junction perovskite solar cells
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Si solar cell Explain why this information is not reported/not relevant.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Enlitech Explain why this information is not reported/not relevant.
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	This is done by Enlitech Explain why this information is not reported/not relevant.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	0.1 cm² Explain why this information is not reported/not relevant.
Variation of the measured short-circuit current density with the mask/aperture area	☒ Yes ☐ No	Supplementary Fig. S122 and Fig. S123 Explain why this information is not reported/not relevant.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	The PCE is certified in SIMIT Explain why this information is not reported/not relevant.
A copy of any certificate(s)	☒ Yes ☐ No	Supplementary Fig. S16 Explain why this information is not reported/not relevant.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	200 cells for spin coated solar cells, 20 cells for solar modules, and 30 cells for blade-coated solar cells Explain why this information is not reported/not relevant.
Statistical analysis of the device performance	☒ Yes ☐ No	Supplementary Fig. S18, S37, S123 Explain why this information is not reported/not relevant.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	The stability test following the ISOS-L-2 protocol was conducted in the nitrogen atmosphere with continuous heating at ~65 °C and under continuous AM 1.5G illumination (100 mW cm⁻²) at open-circuit condition for the unencapsulated devices. The maximum power point (MPP) tracking of the devices was conducted under AM 1.5G illumination (100 mW cm⁻²) at ~60 and 85 °C. The voltage at the MPP was automatically adjusted and applied, and the power output of the devices was tracked over time. Explain why this information is not reported/not relevant.